RABIES VACCINE

Post-exposure — urgent

With RIG for category III

CHEMICAL NAME Inactivated rabies virus (cell-culture vaccines: human diploid cell vaccine HDCV, purified chick embryo cell PCECV, purified Vero cell PVRV)

CLASS Inactivated viral vaccine (active immunisation)

TRADE NAMES Verorab · Rabipur · Imovax Rabies · RabAvert

THERAPEUTIC USES / INDICATIONS

- Post-exposure prophylaxis (PEP) after bites, scratches or licks on broken skin from potentially rabid animals (dogs, cats, bats, foxes, jackals).
- Pre-exposure prophylaxis: veterinarians, laboratory workers, travellers to endemic areas, animal handlers.

ADMINISTRATION

- **IM** into deltoid (adults) or anterolateral thigh (children) — never gluteal.
- **Intradermal** regimens (WHO-approved) save vaccine.
- Thorough wound washing with soap and water for 15 min immediately, then antiseptic.
- Category III exposures: also give rabies immunoglobulin (RIG) infiltrated into and around wound.

DOSAGES

- **PEP (unvaccinated, WHO IM Essen):** 1 dose on days 0, 3, 7 and 14 (4-dose); immunocompromised add day 28.
- **PEP intradermal (IPC):** 2 sites on days 0, 3 and 7.
- **Previously vaccinated:** 1 dose on days 0 and 3 (no RIG).
- **Pre-exposure:** days 0 and 7 (IM or ID).
- Dose is the same for adults and children.

PHARMACOKINETICS

- Neutralising antibodies appear ~7–14 days after starting; RIG provides immediate passive protection until then.

PHARMACODYNAMICS

- Stimulates **rabies virus-neutralising antibodies** → prevents virus reaching the central nervous system (rabies is ~100% fatal once symptomatic).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Injection-site pain, swelling, redness.
- Headache, dizziness, myalgia, fever.
- Nausea.

ADVERSE EFFECTS (SERIOUS)

- Allergic reactions (urticaria, serum-sickness-like with HDCV boosters).
- Neurological events (very rare with cell-culture vaccines).

CONTRAINDICATIONS & CAUTIONS

- No contraindication for post-exposure prophylaxis (rabies is fatal) — including pregnancy and infants.
- Pre-exposure: postpone in acute illness; previous severe allergy.

KEY INTERACTIONS

- **Chloroquine, hydroxychloroquine:** reduce response to intradermal vaccine; use IM.
- **Immunosuppressants, corticosteroids:** reduced response (check antibodies).
- **RIG:** given at different site from vaccine.

NURSING MANAGEMENT

- Immediate wound washing; classify exposure (WHO category I–III).
- Do not suture wound immediately if possible (or after RIG infiltration).
- Start PEP without delay, even if presenting late.
- Give tetanus prophylaxis and antibiotics as needed; observe animal (10 days) where possible.
- Provide written schedule; follow up missed doses.

PATIENT EDUCATION

- Complete all doses on the exact days — rabies is almost always fatal once symptoms start.
- Wash any animal bite or scratch with soap and water for 15 minutes right away.
- Vaccinate pet dogs and cats; avoid stray animals.

KEY POINTS

- Inactivated vaccine; rabies PEP.
- Immediate wound washing 15 min.
- IM days 0, 3, 7, 14 (+ RIG for category III).
- No contraindication to PEP.

SOURCES CHECKED [1006] Verorab, powder and solvent for suspension for injection (UK SmPC)

Full references p. 432–457